ON THE CO-EVOLUTION OF LANGUAGE AND METACOGNITION

Svetlana T. Davidova
https://orcid.org//0000-0001-9190-5937

address for correspondence: svetlana.t.davidova@gmail.com

Abstract

Metacognition is a cognitive property of species of advanced intelligence and is defined as a mental ability to monitor and control one's own mental life. It is a complex cognitive ability, a coordination of various cognitive processes, distributed throughout the brain. Metacognition allows for representation of possible and imagined events, planning , anticipation of possible mistakes in judgement, mental rehearsal etc. It is the cognitive signature to the human mind, which co-evolved with language.
The coordination of metacognition and language is demonstrated differently at different stages of language evolution. At the time of language origins metacognition is mainly a private matter and is rarely expressed in linguistic forms. The emergence of civilization and the invention of writing and its extensive use in science, literature, philosophy encourages attention to one's mental life which becomes the centre of attention and exploration. The two capacities, for metacognition and for language are also coordinated during early ontogenetic development.
The elaborate machinery of language as a culturally developed communicative technology and materialized in writing is the perfect conduit for the verbalization of metacognitive processes by making extensive use of multiple embedding of phrases and sentences.
Metacognitive processes are encoded in linguistic forms differently in different spheres of language use. In spontaneous direct dialogues thought processes are rarely explicitly mentioned as these are most often implied, given that in direct communication the focus is on action and interaction and much less on self-monitoring and self-reflection. In this context evaluating other's mental processes is extrapolated from monitoring overt behaviours.
In sum, two of the most complex human cognitive abilities, metacognition and language are coordinated and intertwined at every stage in bio-cognitive evolution, ontogenetic development and socio-cultural evolution.

Keywords: language evolution**,** metacognition, theory of mind(TOM), oral traditions, writing,

1. Introduction: What is metacognition

Metacognition is a cognitive property of species of advanced intelligence and is a " process by which we monitor and control our own cognitive processes and those of others" (Heyes, 2012, p. 2093). It is the humans ability to " escape form being stimulus -bound and allow self-control of learning and actions" (Metcalf, 2008, abstract). Metacognition is judgement about mental representation, e.g memories, removed from the here-now direct experiences, i.e. it is thinking about thinking, a cognitive ability of the highest order of complexity, introducing a multilevel hierarchy in though. It is the cognitive signature of the human mind. Psychologists distinguish between i*mplicit* and *explicit metacognition*(Frith, 2012). The former is the subconscious and automatic ability to acknowledge other's knowledge and beliefs and incorporate these into one's decision-making to reflect other viewpoints when these differ from one's own, a process taking place during participation in joint activities. Awareness of the interdependence of one's reasoning with the reasoning of others develops very early in infancy and remains present throughout life. *Explicit metacognition* is the ability to purposefully reflect on one's believes and actions, both intended goal-oriented actions and accidental, unintended outcomes, find mistakes, one's and other's, and assume responsibility and/or blame others for unwanted results. Simply put, "metacognitive awareness happens when something goes wrong" i.e. at the point of conflict between expectations and reality, which prompts the mind to question one's knowledge and initiate measures to self-control one's cognitive functions(Metcalfe and Schwartz 2016). In the absence of such conflict the need for questioning one's judgements does not arise. Explicit metacognition begins to develop at about 4 years of age, suggesting a role of experience in learning to reflect on decision-making. A similar differentiation of instinctive and purposeful self-examination is argued by Currithers et all.(2012).

To note, metacognition is a complex ability as it involves the coordination of various components, e.g. perceptual, emotional, cognitive, each with different contributions, which makes it difficult to integrate these in a unified concept and explains why authors differ in their attempts to offer a clear definition. If one identifies metacognition with a mental capacity for monitoring lower level cognitive processes and register lack of knowledge, i.e. implicit metacognition, then various non-human species certainly qualify as self-knowing minds, a function demonstrated also by artificial intelligence. In this sense there is nothing unique to human minds as to metacognitive abilities. On the other hand, Metcalfe, Schwartz (2016) define metacognitive abilities not only as a mere awareness of knowledge(or lack of it), but crucially, as ability to control, with emphasis on control, as a higher level of cognitive complexity, i.e. a crucial distinction between monitoring and registering inadequacy of knowledge, and active supervision and deliberate correction, which implies a mind with self-consciousness. In this much more specifying sense metacognition is a unique cognitive ability, attributed only to human minds. Thus, metacognition is identified as a two-level process, the first attributable to various species and even machines, and a second, built upon the first, attributed exclusively to a conscious mind, capable of self-control and self-correction. It allows for representation of possible and imagined events, planning, anticipation of possible mistakes in judgement, rehearsal etc.

Metacognition is a cognitive activity produced by the association of a number of cognitive abilities, briefly discussed in the following segments.

1.1. Episodic memory

A reflection upon one's mental life presupposes episodic memory, the ability of the mind to recall past events from one's individual experiences and the details accompanying these, e.g. places, participants, smells, sounds, etc., it is mental travel through one's memorable past experiences (Slotnick, 2017). Earlier, anthropocentric perspectives, have attributed episodic memory capacity exclusively to humans. Tulving (2005) defines episodic memory as "conscious awareness of happenings", suggesting the inextricable unity of consciousness and mental travel, its "dependence on remembering "self". Others argue in disagreement that episodic memory is demonstrated by a number of species, e.g. elephants, dolphins, etc. most recently stated by Fugazza et all.(2020) who find a form of episodic memory , i.e. "mental representation of own spontaneous actions and episodic-like memory to recall them" in trained dogs, revealing “ a far more complex representation of key feature of the self than previously attributed to dogs.” (ibid.)

Empirical demonstration of episodic memory in animals is challenged by the absence of language as in the absence of clear articulation of one's mental processes, it can only be detected by mere observation of behaviours, which is insufficient as it is open to interpretation. Additional challenges stem from the fact that interpretations dependent on scholars' preliminary assumptions and variation in the very definition of episodic memory. And because the ability for self-reflection for many is equated with language, a uniquely human trait, self-reflection is viewed as uniquely human as well.

In this sense, a crucial difference between semantic memory, i.e. memory of facts, and episodic memory, i.e. memory of individual experiences and the relevance of this conceptual differentiation from evolutionary perspective is pointed out by Tulving(2005). He argues that semantic memory is ontogenetically and possibly phylogenetically prior and part of the cognitive ancestry we inherited from ancestral species, while episodic memory is understood as a uniquely human evolutionary innovation. Cognitive limitations in animals, are in this context explicable by the fact that they relying only on semantic memory, in the absence of episodic memory. Similar limitations in young children are here explicable with underdeveloped episodic memory e.g. children under 4 years of age have limitations in remembering resent personal experiences and lack the ability to reason about their own mental states.

Significantly, the development of episodic memory i.e. the ability to remember a string of events, is coordinated with development of linguistic skills for narration. In this sense the phylogenetic and ontogenetic development of metacognition is understood in terms of co-evolution with language, which explains its absence in languageless species. Moreover, Tulving (2005) attributes a crucial significance to time travel not only in terms of memories, i.e. a mental travel to the past, but also in imagining future events, so far known to be available to humans only. The uniqueness of human episodic memory, as speculated by Tulving (2005, p.22) was achieved by “ expansion of the subjective time horizon...from the past...into the future”, while animals “ cannot pre-experience possible happenings” (Tulving, ibid.p.40). Thus, although animals and young children do have memories of the past, the ability to imagine the future is found only in human adults.

Episodic memory and self-awareness develops late in childhood, although specific developmental schedule is difficult to discern.

“ there is some considerable period of time, on the order of years, before a considerable shift in thought, allowing the child to remember the past episodically”(Tulving, ibid.p.33).

In other words, a crucial aspect of episodic memory, i.e., imagining future events, or time travel “ constitutes a discontinuity between humans and other animals” (a citation from Studdendorf, Corballis 1997 in Tulving 2005, p.39). The uniqueness of human time travel is in imagining the future and shaping behaviour e.g. by planning, learning etc. to reach desirable outcomes, prevent disasters, i.e. improve survivability. This inherently adaptive value of such mental ability justifies its evolvability.

1. 2. Theory of mind

Humans are highly social species. They are also highly intelligent species. The ability to correctly detect and anticipate the thoughts, desires, aspirations, of fellow humans is known as social intelligence, or " theory of mind". The term was coined by Donald (1993,1999) to mean a specialized cognitive capacity of every normal human which allows him/her to instinctively detect the mental states (beliefs, intentions, etc.) of fellow conspecifics and use this knowledge in joint activities and, especially in communicative interactions, including use of language.This capacity is demonstrated behaviourally in : 1. intentional communication, 2. ability to repair failed communication, 3. ability to teach , 4. to persuade , 5. ability for intentional deception , 6. work on shared plans and goals, 7. share focus of attention, 8. pretend.

ToM is argued to develop in two successive stages, i.e. the ability to detect and infer another's intentions and an ability to make this information explicit by communication (Terrace, 2005). The transformation from the first to the second involves transformation from instinct to intentional behaviour. For humans this is made explicit by communication where language has a crucial role. A form of ToM may have initially evolved as an instinct in some social species and is instrumental in implicit metacognition. It is demonstrated by primates, suggesting that a form of implicit metacognition predates and was a pre-requisite for the evolution of human metacognition. In this sense human metacognition, and especially the ability to exert control over one's decisions, is conditioned not only on knowing one's own mental and emotional states, but also the mental and emotional states of others, given that the individual's mental states and behaviours are intertwined with those of fellow community members (Papaleontiou-Louca 2008). Moreover, to factor others' viewpoints in one's decision-making is necessary as one must evaluate how one's individual strengths and weaknesses contribute to the achievement of common goals and self-regulate accordingly. In this sense " Self-consciousness is a basic motivational tool for getting along with others" (Higgins 2005, p.157), which allows one to self-correct when one's achievements fail to meet others' expectations.

From evolutionary perspective the individual's survival includes adaptation to the environment, which for humans includes social environment as a crucial factor. Self-regulation as adaptation to a shared social reality and self-improvement is the basis of education and language as a tool for instruction and is the basis of human civilization and progress.

1.3. Metacognition in the human brain/mind

As stated above metacognition is a cognitive complex of coordinated components, each with its respective neuronal representations , i.e. metacognition is spatially and computationally distributed within the brain. Metacognitive processes involve multiple locations in the brain as they operate upon basic cognitive functions, e.g. memory, perception, problem solving, coordinated by a higher level cognitive functions of self-reflection and self-control. And such heterogeneous capacities sometimes produce conflicting signals. A typical example of such conflict is the so called tip-of-the tongue experience when a piece of information one is confident to have in the memory storage but is unable to retrieve upon demand. Such conflict of expectations and outcome prompts the activation of anterior cingulate cortex(ACC) implicated in regulating impulse control, decision-making, judgments of learning, as well as in creative problem-solving as a potential resolution of conflict between expectations and reality

(Metcalfe, Schwartz 2016). A damage to ACC is experienced by lack of motivation and apathy. In addition, the hippocampus, implicated in memory retrieval, is said to participate in metacognitive functions as , as per Tulving (2005) hippocampus is the location of both semantic and episodic memory. The prefrontal cortex (PFC) and especially the medial prefrontal cortex(MPFC) and in particular the Brodman area10 (BA10) with the function of control over cognitive processes, especially conscious focus on one's thoughts, is instrumental to metacognitive processing which is why disruptions by lesions in this region result in disruptions in metacognitive processing (Firth, 2012). Interestingly, individual differences in accuracy of metacognitive judgements is correlated with neuro-biological differences of the MPFC) and BA10(Metcalfe, Schwartz 2016). The MPC is crucially the location of convergent processing unifying metacognitive judgements, abstract thought and self-identification in terms of individual idiosyncrasies , e.g. physical characteristics, personality traits. The prefrontal cortex(PFC) is also activated when making judgements about individuals relevant to self-evaluation. So PFC and particularly BA10 is the location of metacognitive functions coordinating " just about everything having to do with the self"(Metcalfe, Schwartz 2016, p.30)

2. Evolution of metacognition and human speciation

The evolution of complex cognitive traits in humans is difficult to determine and is understandably a focus of debates, usually clustered around two alternative visions. On the one hand cognitive evolution in humans is envisioned in terms of gradually increasing sophistication of pre-human predispositions, i.e. a continuity argument, or, alternatively, key human cognitive traits are viewed as "a leap forward" by which a cognitive Rubicon marks human cognitive uniqueness and exceptionality.

2.1. Evolution of metacognition, the continuity vs. discontinuity arguments

The continuity vs. discontinuity debate over the question of human cognitive evolution is extended in the search for the evolutionary history of metacognition. Tulving (2005 and elsewhere) is a prominent proponent of the discontinuity argument, pointing at metacognition as a " missing link", a position justified by significant cognitive disparity in metacognitive abilities between human and non-human species.

And although the cognitive differences between us and non-human species are obvious and indisputable, this does not suggest lack of continuity as, as Kinsbourne (2005) explains " continuity does not imply linearity"(ibid. p.144). And as a general tendency in evolution, despite significant gaps in linearity, adaptive traits still form consecutive stages along a continuum across species. Applying the same train of thought to cognitive abilities Kinsbourne regards "unconscious, conscious and self-conscious states not as discrete but as differing only in degree on a continuum of differentiation of neural activation pattern...(in a brain) " (ibid.p.151). Carruthers et all.(2012) also offer a continuity perspective by postulating that metacognition, or self- control over one's mental life and decision-making, has evolved from theory of mind, a cognitive capacity demonstrated by primates as highly social animals, purported to have evolved in humans into self-knowledge by " turning our evolved mind-reading capacities on ourselves"(Carruthers et all.2012, p.16). In other words the significant gaps in metacognitive abilities between humans and the rest of species does not necessarily signal discontinuity as adaptive traits rarely evolve following unbroken chain of ever increasing functionality.

From a slightly different perspective Nelson (2005) espouses a continuity argument and finds parallels between ontogenetic and phylogenetic development of complex capacities, e.g. metacognition, in terms of successive stages of self-organization. To remind, self-organization is an event or a succession of events, by which a novel complex entity is formed with unique characteristics by coordinating multiple individual components which act in concert. In cognition complex cognitive entities emerge from a combination of pre-existing bio-cognitive formations, recruited for new roles within a new cognitive organization. This process of harmonizing the functions of multiple cognitive traits into a novel capacity, is outlined by Donald (1991 and elsewhere) who explains the formation of novel and unique cognitive abilities in humans, e.g. metacognition, and its application in deliberate individual improvement by practicing and instruction, as having roots in earlier simpler abilities in apes and primates.

In the field of evolutionary linguistics the discontinuity argument for human cognitive traits is consistent with the biolinguistic argument for the modular nature of language processing, arguably concentrated in a language bio-program, or Universal Grammar, proposed by Chomsky, Bickerton and fellow likeminded scholars, a position articulated over various decades in multiple publications, too numerous to cite here.

That said, a complex cognitive trait is usually a convergence and interplay of a number of simpler traits, each with different evolutionary history, which speaks against the discontinuity argument. Moreover, extensive literature on comparative neuroanatomy of human and non-human brains has effectively debunked the discontinuity argument in evolutionary linguistics, demonstrating clear continuity in both cognition and communication, pointing at various degrees of language-relevant properties in a number of species (Fitch 2009, Zuberbuhler, 2015, to name just a new).

2.2. Degrees of metacognition

To note, channeling one's activities in response to changing environment is an ability available to numerous species of animals and even plants, e.g. plants adjust their movements in reaction to the sun and animals make choices on strategies of escape from predators. Thus, reactionary adjustments of behaviour are nothing new in the living world, but this capacity must not be mistaken for metacognition.

The ability to anticipate events and their consequences and adapt one's actions accordingly in order to assure one's protection rather than merely responding to events of life has a clear survival advantage, as it signals transformation from mere recipient of acts of nature to active participation in shaping one's faith. In this sense, given our cognitive continuity with other primates, it is pertinent to suspect some form of metacognition to be available to some primates, most likely great apes, e.g. a form of metacognitive abilities by representation of counterfactuals is detected in monkeys (Metcalfe 2008). Tomasello (2023) argues that a fairly sophisticated social intelligence, or theory of mind in apes has co-evolved with metacognition as apes monitor behaviours and make inferences about conspecifics' knowledge and use it to predict and anticipate future behaviours, i.e. they demonstrate knowledge about knowledge. Chimpanzees demonstrate understanding of goals, believes, plans. And crucially, it is observed that “ The chimpanzees understood not only what the experimenter was trying to do, his goal, but also the rationality behind the choice of action plan toward the goal”(Tomasello, 2023, p.30.) Very similar abilities are demonstrated by bonobo and orangutans. Importantly, apes seem to know when they have insufficient information for their decision-making and aim to improve by seeking additional information, i.e. apes are capable of “ self-monitoring and controlling their decision-making”(Tomasello, 2023, p.32).Thus, the ape mind is capable of “second -order tier executive functioning-metacognitive or reflective functioning” (Tomasello, ibid. p. 32). See also O 'Madagain et all. 2022). Moreover, by understanding of their own mental states apes extrapolate understanding of mental states of others (Kornell 2009). Importantly, studies which aim to uncover potential metacognitive abilities in lower animals, summarized by Qu et all. (2023) demonstrate information-seeking and uncertainty in decision-making, abilities vital for correcting errors in behaviour and crucial for survival, especially in changing environments as animal minds make judgments on objects and events. That said, animal minds lack self-reflection abilities, typical for humans. In this sense it is noteworthy to mention that the hippocampus, the location of memory storage and retrieval, is a brain region with anatomic features “ highly conserved across rats, cats and monkeys” (Slotnick 2017) e.g. “ The elephant hippocampus is of similar complexity to humans”(ibid.) In short, a form of metacognition, albeit primitive, is available to primates and even lower animals, demonstrating cognitive continuity.

Demonstration of continuity, nevertheless, does not negate the uniqueness of human metacognition. To reiterate, metacognition is not simply judgement and differential reactions to current experiences, but, crucially, judgement about mental representation, removed from direct experience, thus, thinking about thinking, a cognitive ability of the highest order of complexity, introducing a multilevel hierarchy in thought and, as per our current state of knowledge, available only to humans (Metcalfe 2008). In essence “ this human ability seems to differ qualitatively, rather than quantitatively from that seen in other animals”(Firth, 2012, p. 2220).

Significant, from evolutionary perspective, is the finding that during human evolution BA10, associated with metacognitive processing, has undergone the most significant enlargement and changes in internal reorganization, as compared to other brain regions (Firth 2012). In addition, given that metacognition is a composite property, incorporating a number of pre-existing cognitive components, it is logical to suggest that each of them evolved from some rudimentary level and became interconnected at some stage in primate evolution. In this sense demonstrations of joint attention and shared gaze in young children and trained apes suggests earlier evolutionary stages of theory of mind(see Tomasello, 2023 and elsewhere; Terrace, chap.3 in Terrace, Metcalfe 2005). That said, although advanced metacognitive abilities may have some innate component, given the argument for cognitive continuity, in humans these are largely emergent starting in early development by training and become continuously elaborated into adulthood, i.e. human metacognition is teachable.

To sum up, the studies referenced above demonstrate **a.** that metacognition must not be regarded in binary terms as present or absent but as graded cognitive ability, available to various species to various degrees. **b.** it is a cognitive attribute crucial for survival in natural habitats and in social environments, which suggests that it could have been evolutionary target.

2.3. Human metacognition and cultural diversity

Given that some, albeit primitive, form of metacognition is suggestive in non-human animals and given the continuity argument, one would expect metacognitive abilities in humans to have perhaps genetic, underpinnings, inherited from ancestral species. In this sense Nisbett, Norenzayan (2002) suggests that human cognitive processes are influenced by innate, and , thus universal, bio-cognitive architecture of the brain and its functioning. This bio-cognitive inheritance constrains human reasoning by placing limitations " on the range of beliefs and forms of expression that can be fond in different cultures" i.e." not just any beliefs, or ways of constructing the world are possible for humans"(ibid.) For example, people universally categorize reality, material and social, in terms of dichotomies, e.g. good/evil, male/female, left/right, etc., indicative that some ideas are " easier to think and communicate", as they fit with the " innate contours" of the human mind. This explains the observable regularities in the ways all humans perceive and organize knowledge of the world, demonstrated in commonalities in folk theories and religious belief systems. Moreover, human infants demonstrate " number sense", suggesting innate predispositions to conceptualize reality in quantities. Thus, at the foundation of all human reasoning, from tribal belief systems to West -European and North American philosophy and science, are " folk taxonomies", emphasizing that cultural diversity is rooted in common cognitive foundations.

At the same time, Nisbett and Norenzayan(ibid.) find that culture influences and shapes human cognition as crucial differences in reasoning patterns are found in traditional and industrialized societies. For example in traditional, non-literate societies knowledge is extracted from intuitions, concrete experiences and direct interactions with material objects, while literate minds in industrialized cultures have preference for formalization of knowledge based on logical reasoning, distanced from perceptual experiences. These differences in styles of reasoning are instilled in early childhood, emphasizing the role of education and training and reflected in language. In short, systems of reasoning are responsive to cultural variation. Moreover, socioeconomic systems of organization shape the systems of thought, while at the same time changes in social organization and cultural values direct changes in the organization of knowledge and reasoning. In essence, species-specific innate predispositions form the foundations of human reasoning and on these bio-cognitive foundations cultures shape the innate human potential into diverse systems of thought to fit their respective values and philosophies. Thus, nature and nurture interact as two complementary factors in the formation of human minds.

Cultural differences deeply impact complex cognitive processes, specifically metacognition. In recent studies Heyes et all.(2020) find limited participation of genetic factors and suggest crucial influence of culture and learning in metacognitive processes. These influences are especially relevant as they affect the ontogenetic development of explicit metacognition in coordination with culture and in adaptation to cultural changes. Explicit metacognition, i.e. deliberate evaluation of memories and past decisions, is time- consuming. It makes possible recognition of errors in decision making, alter cognitive demands to avoid cognitive overload, etc., e.g. children and adults can regulate and adapt learning patterns. Moreover, metacognition allows for sharing of individual's mental states, crucial in enhancing collaboration in group decision making. And given that explicit metacognition develops late in childhood, it is reasonable to speculate, after Heyes et all. (2020) for a crucial role of learning within a socio-cultural context.

From evolutionary perspective strong influence of cultural evolution, and not a product of gene-based selection, are more likely, given that differences in metacognitive strategies, arguably, have evolved to adapt to different cultural environments. As an example, it is noticed that " students form western, individualistic cultures(USA, Australia, New Zealand) express more confidence in their decisions, than students from East Asian group-oriented cultures(Japan, Hong Kong, Taiwan). Consistent with these findings are brain imaging studies which show that " The medial prefrontal cortex, a focal area for metacognition, is more strongly activated during self-assessment tasks in westerners than in people from less individualistic cultures"(Heyes et all. 2020, p.357). Moreover, a role for culture and especially for education and training is suggested by individual differences in metacognitive development, created by parental guidance and teacher's instructions, encouraging youngsters to develop self-confidence, most effectively by using verbal stimulation with phrases, e.g. "You can do that.", stimulate evaluation of one's decision-making process by using phrases, e.g Are you sure?, etc. In sum , although implicit metacognition is likely to be explicable with some form of instinctive predispositions, especially reliant on theory of mind as a human instinct, explicit metacognition is much more likely to rely on socio-cultural stimulation, e.g education and training and thus, subjected to evolutionary processes in cultural contexts.

3. Articulation of metacognition in language

It has been established that metacognition and language, two unique human abilities, are interconnected. It has also been established that metacognition is a multicomponent cognitive property where each individual component interacts with linguistic abilities in different ways. For example, theory of mind, a vital ingredient of metacognition, is intimately connected with communication, given that both develop in tandem ontogenetically and impairment in one parallels impairment in the other, e.g. autistic individuals have difficulties both with theory of mind and with language. In this sense Oesch and Dunbar (2016) find strong correlation between language development in children and development of theory of mind. In addition, a number of scholars, from Aristotle to Tulving, have argued that the uniqueness of human memory recall as " true episodic memory" (see summary in Menzel 2005, p.191-) resides in its externalization by language. And as per Oesch and Dunbar (2016) " mentalizing capacity"or Theory of Mind provides the natural connection between recursive thinking and communication, exemplified in the use of verbs which require sentential embedding , e.g. *believe, think, suppose, imagine, guess,* etc. Significantly, it is argued that the very concept of self , i.e. the sense of having individual identity, responsibility and authorship of one's behaviour and its implications, is tied to its explicit articulation in language. In addition, language influences one's reflection upon oneself, makes it easier to identify one's mental states, define clearly individual goals, share information and agree on joint future projects. In this sense it has been argued that the properties of language systems influence one's thought process in decision-making , including one's self-reflection, which is why, although a number of scholars have attributed episodic memory to primates and even lower mammals (Menzel 2005), assessing the mental life of languageless creatures is difficult.

Notably, it has been determined that verbalization of mental processes is different in spoken and written languages, to be discussed in some detail in the next segments.

3. 1. Metacognition in oral traditions

Language influences thought and spoken and written languages influence metacognition and shape young developing minds in vastly different ways.

Speech is universally the dominant venue for realization of the language system and influences the way language is perceived, processed and learned. Moreover, human thought processes, including human consciousness are shaped by spoken language throughout human history.

“ Speech is inseparable from our consciousness and it has fascinated human beings, elicited serious reflection about itself, from the very early stages of consciousness, long before writing came into existence” (Ong, 2002, p.9)

Speech shapes the human mind in a very particular way, illustrated by the oral traditions, e.g. myths, tales, legends of preliterate peoples. For example, individual linguistic forms are perceived and interpreted as a stream of sounds, not as individual entities with clear margins as dictated by writing. Moreover, the spoken word is a reflection of the nature of sounds, e.g. each time a word is uttered is a momentous occurrence, a unique event of emitting a signal, after which it evaporates and leaving no trace. In addition, the stream of speech is perceived as continuous, uninterrupted stream of sounds, which stimulates the perception of reality, e.g. space, time, as continuous. Linguistic forms have no fixed meanings as each event of use imposes different interpretation within a context, while in writing they are fixed labels. Notably, the word choice in the poems is determined less by meanings and primarily by the “ metrical needs of the passage”, composed in hexameter, with abundant use of epithets, suggesting oral poetry is meant not to inform but to impress and excite. Moreover, the story is composed anew at each performance, i.e. it is “ stitched together differently at each telling”(ibid. p.22). Oral traditions are composed following established patterns of thought and formulaic language with liberal use of proverbs and perpetuated by exact repetition to assure their preservation in the absence of writing. Significantly, words in oral traditions are instruments of action, not instruments of thought. In the minds of non-literate people they project magical power, while in written language they are objects with visual boundaries.

These markedly different ways of perceiving and processing linguistic forms inevitably shape thought. Thought in oral tradition is perishable and is preserved only by memorization. The story in oral traditions is a vehicle for organizing, preserving and disseminating knowledge, and because of that it must be impressive to be memorable. This is why legends, tales and myths of creation are designed around impactful events, where the characters are heroic figures, who represent communal values of good and evil and their struggle for dominance. Oral narrations are also shaped by the innate affinities of the mind to remember themes with strong emotional impact, designed to instil pride of communal unity and sense of belonging. In addition myths, legends and other expressions of folk wisdom are organized to take advantage of the mind's affinity to memorize sound patterns, which is why their style is rich in rhythmic repetitions, proverbs and epithets, with traditional themes featuring warriors and heroes in epic battles, where the fight of good and evil serves as scaffolding for organizing communal knowledge. Patterns of reasoning are standardly expressed in proverbs, whose repetition instils moral behaviour and wisdom. In this way knowledge and wisdom of generations past is perpetuated as cultural conventions to assure preservation of values and stability in a society. At the same time it “establishes a highly traditionalist or conservative set of mind that inhibits intellectual experimentation.”(Ong, 2002, p.41).

Some of the best examples of oral tradition are the Greek poems Iliad and Odyssey, authored by Homer, who, as per renowned scholars with knowledge of the subject, referenced by Ong (2002) was illiterate, and whose verse illustrates the properties of popular language and the illiterate mind, rather than the mind of a learned author.

Structurally oral traditions display strong preference for parataxis, where complex thoughts are verbalized by stringing phrases and sentences, as opposed to hypotaxis, i.e. extensive use of clausal and sentential subordination. Two versions of the same narrative from the Genesis illustrate the structural differences between oral and written verbalization of the same thoughts. The oral version:

“In the beginning God created heaven and earth. And the earth was void and empty, and darkness was upon the face of the deep; and the spirit of God mover over the waters. And God said: Be light made. And light was made. And God saw the light and it was good; and he divided the light from the darkness. And he called the light Day, and the darkness Night, and there was evening and morning one day.”

The written version:

“ In the beginning, when God created the heavens and the earth, the earth was a formless wasteland, and darkness covered the abyss, while a mighty wind swept over the waters. Then God said “ Let there be light” and there was light. God saw how good the light was. God then separated the light from the darkness. God called the light “day” and the darkness he called “night”. Thus evening came and morning followed-the first day.” (Ong, ibid.p.37).

Elements of oral language survive in children's fairytales in characters with extraordinary lives, e.g Cinderella, Red Riding Hood, the Three Little Pigs and evil characters, e.g. the wicked wolf, the sinister stepmother, etc. To note, redundancy and rhythmic repetition are techniques borrowed from oral traditions and used in public speaking by literate minds.

In short, orality shapes the mind by introducing biases which stimulate emotional expression vs. rationality and linguistic tradition vs. creativity.

3.1.1. Orality and the structure of conversations

Oral traditions are usually told in monologues, but we also discuss with others our mental states in conversations and the structure of the conversation and the extralinguistic and cultural circumstances around it also influence the way we share private information about one's mental and emotional life.

A conversation has a distinct organization, it is a flexible unit of interaction with a distinct meaning and structure and its own “ communication management” (Ginsburg, Poesio, 2016, p. 14). A common set of assumptions shared by the participants, i.e. initial common ground at the start, often undergoes alteration as the conversation progresses as with each move a new repository of facts is added to the initial common ground. A dialogue usually begins with a greeting word or a phrase. ex. Hi. Hello. Some languages have also phrases used exclusively as a response to greeting, where greetings are usually about wishing god's blessings for good health.

ex. A. God give you health.
B. God helthify you (Ginsburg, Poesio, 2016)

The initial topic of the conversation can change in the process as some communicative actions are clarifications of earlier statements and others are not directly related to immediately preceding moves.

ex. A. Several people showed up today.
B.Who? (Ginsburg, J. 2001)

The extralinguistic context determines the range of topics and the semantic and grammatical organization of the dialogue, e.g. a conversation in a bakery is different from a conversation in a courtroom. Non-linguistic signals, e.g. a head shake and nod, laughter, eye rolling, etc. accompany and complete the linguistic message . Silence, i.e., the absence of explicit reaction, often carries a meaning. A dialogue is described as a type of game, where every move triggers a countermove. Examples can be found in Ginsburg 2001, Ginsburg 2008, Ginsburg Poesio, 2014). In direct spoken dialogues participants report on direct observation of events and sharing of recent experiences is highly prioritized before reflection.

ex. There is a rabbit in the grass. My friend is returning from hunt. The antelope is running away. It is raining.
Piraha: I only want some fish.(Everett, 2005, p.623)
I am not afraid of beings with blood.(Everett, 2005, p. 625) (translated by Everett).

To note, Piraha language does not have a verb for *THINK*.
In short, expression of orality in dialogues is conducive to interactive experience, not of introspection and self-analysis.

3.1.2. The language of spontaneous face-to-face dialogues

As underscored by usage-based perspectives the language system is tailored to fit language use and language is used primarily in communication, where the direct dialogue is the primary form of language use. And as per the theory of communication language in its use in dialogues is a case of inferential communication system with the following distinctive properties:
a. The inferential system has information-based, not structure-based internal organization, that is, organized around information structure (topic vs. focus)
b. It exist mainly in spoken form, where intonation assumes some grammatical functions, e.g. the formation of questions without the use of question words.
c. The building blocks of the system are flexible associations of form and meaning as standard meanings are interpreted with context-dependent flexibility.
d. These form utterances composed of the most frugal use of linguistics constructions, the choice of which varies with the circumstances.

e. Use of fixed expressions, e.g.“ when all is said and done”, “ when pigs fly”, and proverbs e.g. “ You snooze, you loose.” is common. The meaning of an utterance in a dialogue is represented by a variety of linguistic forms , from complete sentences to non-sentential utterances. Most verbs have incomplete argument structure with only a single argument which have meaning only in a specific context.
ex. That one.
When an utterance is a full sentence the order of the elements is flexible to signal speaker's attitude.
f. The meaning of a sentence is different from the meaning of the utterance and the difference between the two cannot be stipulated in advance. The meaning of an utterance is calculated as the meaning of a sentence and the speaker's communicative intentions. Utterances communicate the intended meaning in addition to the speaker's attitudes, so meaning is largely situated in context.
g. The meaning intended by the sender is most often different from the meaning understood by the receiver. An inferential system is based on the assumption that participants are individualities with different minds and different life experiences in different communicative circumstances , which creates the potential for different interpretations of the same linguistic forms by individuals of different backgrounds.
e. Oral language reflects the idiosyncrasies of local cultures, while written texts are meant to have a single, correct and timeless interpretation, e.g. articles and books on scientific discoveries from centuries ago have the same meanings today.

The properties of the dialogue influence its cognitive processing. An utterance is processed with high speed , e.g. linguistic item is processed on average as follows: 65 milliseconds (msec) for the processing of a phonological form, 250 (msec) for processing the semantics, 1-2 sec for processing the grammatical properties of a sentence (T.Givon ,2002, p.74). The high speed of processing of speech articulation and perception exerts influence on the flow of thought as “...an intonation unit can express no more than one new idea. In other words thought articulated by language within the circumstances of a dialogue proceeds in terms of one such activation at a time, and each activation applies to a single referent, event, state, but not to more than one” (M. Mithun, citing W.Chafe 1994, in T.Givon , M.Shabatani 2009, p. 67).

In this sense given that self-reflection and self-corrective thought is a cognitive process requiring time and isolation away from the constant monitoring of the the rhythm of a conversation, where the focus is on the moves of the participants in the “ game”, the focus is on the here-now, circumstances hardly conducive to critical self-examination. That said, the conversation setting allows for some form of introspection and self-analysis.

ex. “ I'm just really anxious, not anxious, anxious is the wrong word, I am excited about tomorrow. (Ginsburg, 2016)

Notably, close attention to extralinguistic behaviour, e.g. nervous gesticulations, facial expressions, etc., e.g. signals of self-doubt, self-confidence, which unintentionally reveal emotional and mental states of conversation partners, become indicators of their metacognitive functions and factored in adjusting one's next moves in the game of dialogue.

In sum, orality in folk narratives and in direct face-to-face dialogue is hardly the behavioural frame conducive to reflection and self-evaluation, where the function of language as action maintains constant connection with reality, greatly reducing one's opportunities for introspection.

3.2. Metacognition in written texts

Writing creates a new structure for the narrative as it establishes distance between the story and the context. The story assumes a life of its own and its verbalization becomes much more dependent on language, which explains the extensive use of complex grammar, e.g. phrase and sentence embedding as a compensation for lack of social and cultural context. And while in oral traditions thought is " redundant and copious", writing allows for analytical examination and creativity. Writing encodes knowledge in abstract categories, which removes the reader from direct experience and converts it into experience of the mind. Thus writing stimulates inward looking and dissection of one's mental life and decision-making, orality stimulates outward emotional attitude towards direct experiences evaluated through the lens of communal integrity. Moreover, while in oral language communicators are real people with distinct personalities, conversing in real time at real places, the potential audience is unknown to the writer and is in a sense abstract.

" The writer's audience is always a fiction...The reader must also fictionalize the writer"(Ong, 2002, p.99).

Writing creates distance between author and audience, opening space for introspection, self-analysis and, perhaps self-criticism and individual improvement in reasoning and behaviour for the individual mind. The focus on mental life is instrumental in the formation of civilization, where the socio-economic environment creates division of labour and the appearance of professional thinkers and writers creates the conditions for cultural growth with the emergence of religion and philosophy.

3.2.1. Language as a code and the technology of writing

In current linguistic theorizing language is defined in terms of computation by borrowing concepts from mathematics and Turing's theory of computation, where a finite and predetermined set of abstract symbols automatically combine according to a finite number of equally predetermined rules, producing infinite number of combinations. In this context natural language is understood in terms of artificial systems as a technology on par with computer software, i.e. a cultural invention.

" As with other symbolic systems that encode logical information, such as arithmetic, logic and computer programming, it is essential to get the parentheses right, and that's what phrase structure in grammar does" (Pinker, 2003, p.18)

As a type of software, the language system is best described as a code and most accurately articulated in writing as it answers the communicative demands of civilization for detailed exposition of complex ideas to potential audiences at a distance of space and time, which justifies the following defining features:

a. It is composed of linguistic primitives, understood as discrete, object-like abstract entities which stand in fixed relations with one another and have existence independent of their users.
b. These consist of equally discrete and finite component parts. So sentences are decomposed into clauses, phrases, words, morphemes, syllables, phonemes, phonological segments.
c. Members of the lexicon are one-to-one stable associations of a meaning and a form, i.e. synonymy and homonymy is non-existent. These are defined by their membership in discrete and well defined grammatical categories and organized into sentences according to predetermined and fixed principles of grammaticality.
d. The meaning of a sentence is the sum total of the meanings of the composing words and their place in the architecture of the sentence. In this sense meaning is in the language itself.
e. A sentence is the encoding of a complete thought. Explicit and complete mapping between semantic structure and grammar is the norm. All thematic roles in the theta grid of a verb are expressed in grammatical categories. The agent consistently occupies the subject position in the sentence structure.
f. The language system is self-contained, stands alone, independent of context of use. This facilitates the uniform decoding of the meaning by people with vastly different experiences and views at any place and time , creating distance between communicators.
g. The sentence structure is highly detailed , it contains multiple embedding of phrases and sentences and highly abstract grammatical forms.
h. The message for the sender and the receiver are identical.
i. The function of code systems is mainly to inform, ergo, sentences are mostly statements. If delivered in speech they are usually articulated in monologues although in the majority of cases if delivered in a form of speeches, lectures, etc. these are first realized in writing and the oral presentation is by reading from a script.
J. The writing technology enables the exposition of the language code in discrete and visibly distinguishable basic units, the letters of the alphabet or other forms of writing, which are spatially organized, static and atemporal units. This encourages thinking with " analytic exactitude" (Ong, ibid.p.102)

4. Orality and literacy influence human mind, brain and language

Oral narratives and written texts shape human cognition differently. Oral traditions shape human knowledge by relying on reference to observable facts of immediate experiences, a result of a close contact with the natural world, while writing shapes knowledge in terms of abstract categories, removed from direct experiences. In oral cultures knowledge is derived from practical exposure, observation and intuition and perpetuated by apprenticeship of practical skills, while literacy allows for knowledge preservation by explicit instructions and teaching, which demands abstract concepts. Significantly, in oral cultures knowledge means confirmation of the communal understanding of reality, i.e. "communal identification of the known"(Ong, 2002, p.45), thus knowledge is part of communal identity, while writing creates distance between the subject and object of knowledge, between the knowledge seeker and reality, allowing for objectivity of findings and creativity in verbalizing knowledge in linguistic forms.

The formulaic nature of oral poetic traditions, repeated over generations and over millennia, becomes engraved deep into the mind and perseveres even upon acquaintance with the written word, probably because, as per Ong(2002) redundancy and repetition is a natural fit to the functioning of the brain, instilled by evolution, while writing is a technology, a very recent development in human evolutionary history and product of civilization, foreign to brain's functions.

Writing is an activity of the individual mind, removed from direct human contact, which allows time for reflection and rational evaluation, encouraging metacognition. In contrast the oral mind identifies the individual with the community and is not interested in self-reflection. This distancing of the mind from reality and from communal identity encourages the awareness of the self and creates conditions for focusing on one's mental life, i.e. metacognition is facilitated by literacy.

“ Oral communication unites people in groups. Writing and reading are solitary activities that through the psych back on itself.”(Ong, ibid.p.67).

In cultural environment based on literacy knowledge is obtained by applying the principles of logic by deductive reasoning, invented in ancient Greece, which allows for verification and refutation of hypotheses, made possible by writing. In comparison, in oral cultures the communal believes are irrefutable, as they are not subjected to verification. In short, written literature fosters reflection, oral stories depict communal life and values as action.

To reiterate, writing influences the perception of linguistic forms. For example the individual word is perceived differently in orality and in writing, e.g. written words are perceived as separated by empty spaces on the page, which conditions the mind to conceive words as discrete entities. In contrast, in oral language a word is perceived holistically, inseparable from the stream of speech within a rhythmic phrase. Thus writing stimulates inward looking and dissection of one's mental life and rationality in decision-making, orality stimulates decision-making based on perceptual experiences and traditional belief systems. Moreover, in oral cultures words are intertwined with somatic experiences, i.e. “ words and objects are never totally distinct”(Ong, ibid.p.66). It is also suggested that writing introduces unnatural patterns to the brain “preventing expression(of thought) from falling into its more natural patterns (reflected in oral traditions, emphases added) (Ong, 2002.,p.39). In written languages verbalization of mental activity is facilitated by abstract vocabulary, e.g. nouns *idea, analysis, judgement, examination, inference*, etc. the frequent use of verbs e.g. *reflect, assume, understand,* etc. the argument structure of which demands the use of complex grammatical structures, which requires departure from the physically active life and demands free time and intense mental concentration. The verb *think* and its synonyms in literate languages are the most common tools for verbalization of metacognition. To note, Cambridge Dictionary has about 50 synonyms of the English verb *think,* and the website synonimosonline.com lists at east 26 synonyms for the Spanish verb *pensar,* demonstrating the multiplicity of options written languages offer for facilitating and encouraging verbalization of metacognition. And *think* is a transitive verb which, as all its synonyms, demands a sentence as its object, thus, it demands sentence embedding, potentially allowing for unlimited recursion. The same structural requirements apply for the verb's translational equivalents in European languages. Overall, there is significant difference between the original language system, illustrated by the spoken word, and its transformation imposed by the written word.

Importantly, literacy affects cognitive activities, i.e. the thinking and learning processes and speaking style, e.g. it is not difficult to notice that highly educated individuals speak in paragraphs (see Ong 2002 for details). It follows that writing, as a communicative technology, shapes the human mind.

“ Technologies are not mere exterior aids but also interior transformations of consciousness and never more than when they affect the word”(Ong, 2002, p.81)

And because writing and reading are activities of the individual in private , this encourages the focus of the mind on itself, its individuality, e.g. self-analysis, self-reflection, self-determination, individual growth.

“ Writing heightens consciousness” (ibid.p.81)

In fact, thought shaped by writing, is quite different than thought shaped orally. It introduces a structure quite different from that of the dialogue, e.g. title, introduction, paragraphs, final thoughts, distributed on pages and devoted to a single unifying topic. Literacy is suggested to influence the structure of one's memory, e.g. illiterate individuals' memory storage contains more concrete examples, whereas literate people organize their memories by using abstractions.

Literacy is found to influence not only cognition but, importantly, brain performance and connectivity. Ardila et all.(2010) argue that exposure to literacy and education influences brain activity and results in changes in brain connectivity and potentially brain lateralization. For example fMRI tests show increased activation of the left hemisphere in brains of individuals with post secondary education, compared to illiterate individuals where the right hemisphere is more engaged. In addition, exposure to literacy and education is argued to stimulate executive functions by directing focus on linguistic performance and making it an object of reflection and judgment.

And although writing as technology is by definition unnatural, it becomes “ naturalized” by altering the brain's activity and becomes essential to human life. So paradoxically, as pointed out by Ong(2002) “ artificiality is natural to human beings.”(ibid. p.81). This informs theories of language which identify language as a system by its use in written texts, where focus is on structure and examples of language use are analyzed in terms of dichotomies,e.g. correct vs. incorrect , as compared to the standards of correctness determined by written language, while the language of oral dialogues is seen as a unstructured concoction of linguistic forms.

To note, recently a number of studies, popularized on www.healthline.com, and elsewhere, point at wide-ranging positive effects of writing and reading on the brain health by reducing stress, enhancing learning, improving memory, reducing the effects of aging and improving overall psychological health. This is not to say that minds conditioned by oral traditions are less intelligent or somehow mentally deficient, as difference does not imply inferiority, it is just difference.

5. The co-evolution of language and metacognition

To remind, defining language as computation, i.e. manipulation of symbols, although conceived as a brain-internal processing, irrespective of its usability in communication, fails to acknowledge that symbols by definition are signs and signs presuppose communication, i.e.

exchange of messages between two communicating parties, a sender and a receiver. In this sense the neuronal activity within the human mind by which signals are transmitted throughout the brain, an activity taking place during metacognition, is in essence communication.
On the other hand symbols are signs based on agreement between communicating parties, i.e. they exist only by social contract, which presupposes a form of social organization. In this sense the use of language as a symbolic system for silent reflection within the individual mind, naturally predisposed to sharing one's mental life with fellow minds, is a misrepresentation of both symbols and human minds, as in my mind the neuronal activity within the mind cannot be described as language.

Significantly, sharing one's mental life presupposes ability for understanding other's minds, e.g. experiences, aspirations, desires, and factoring these into one's decision-making and/or manipulating other minds to fit one's own aspirations. In short, language use is based on a theory of mind.

In this sense it is prudent to assume that a form of theory of mind has predated language, an assumption supported by the demonstrable abilities of languageless primates and human infants to engage in joint activities and collaborate in problem-solving tasks. The mind-reading abilities of modern primates may give a clue of earliest stages of a human theory of mind which includes ability to follow gaze, to guess intentions of others, etc. From this initial and rudimentary form of theory of mind Givon, Malle (2002) envisions a co-evolutionary scenario where primitive states of both language and theory of mind co-evolved and mutually encouraged each other's further evolution, i.e. "one of the faculties emerged first in a primitive form and facilitated a primitive form of the other, and soon the two plunged into a race in which advances in one faculty repeatedly enabled or demanded advances of the other. What we have here is a version of the arms-race argument. " (Givon, Malle, ibid., p. 274). A similar argument for a form of theory of mind having evolved initially for silent reflection, one's and others', as a predisposition for the evolution of language is made by Terrace(2005).

To remind, the formation of a primitive lexicon and its implementation in a lexical protolanguage (Bickerton 1990; Jackendoff, Wittenberg 2014) as initial stage of language evolution, presupposes the ability to form conventions by mutual agreement, based on a theory of mind. The initial function of (proto)language, judging by its use by young children and language-trained apes, was referential, i.e. referring to objects and actions in the material reality. And it is reasonable to suppose that at some point it has become adaptive for individuals to pay attention to one's own mental processes and states and infer and influence mental states of other minds, and crucially, to communicate by referring to mental states and processes, via using linguistic forms. As it has been demonstrated by dialogues, this does not necessarily demand the use of complex grammar.

5.1. Co-evolution of the brain/mind and metacognition

The transformation from pre-human communication, termed as protolanguage, into fully modern language by the introduction of highly abstract grammatical concepts and forms is envisioned by the biolinguistic approach in terms of genetic transformation expressed phenotypically by transformation of the pre-human brain into a human brain with the appearance of a bio-cognitive module, i.e. Universal Grammar(UG), computationally specified for processing complex grammar. And given that the computations of UG produce a linguistic code, in biolinguistic context the language code is argued to be a product of biological evolution. Moreover, the biolinguistic perspective has consistently argued that the original function of the language algorithm has been the hierarchical organization of the individual's reasoning as Language of Thought, although it is not clear if metacognition is part of it.

That said, the only empirical demonstration of the language code is by writing dated at 6,000 years ago. In addition, the biolinguistic argument for the evolution of UG some 100,000 years ago is a speculation, given that for the vast majority of time since its origination language has been used in dialogues, which, as illustrated above, has internal organization deviating from the language code.

A more plausible argument by Pinker Bloom(1990) hypothesizes that language has originated as a cultural invention, triggering some adaptation by evolving bio-cognitive properties via Baldwinian evolution. To remind, Baldwinian evolution happens when cultural behaviours and practices, initially learned for a period long enough to become stable part of the environment, trigger adaptation by converting some aspects of learning into innate predispositions, thus decreasing to a minimum and even eliminating learning time and effort, i.e. in essence Baldinian processes convert nurture into nature (see Jablonka ,Lamb 2005 for detailed discussion). And although Pinker and Bloom (ibid.) invoke Baldwinian processes to explain the evolution of the full package of UG, it is more likely that internalization of previously learned behaviours was limited to some crucial aspects of language, e.g. speech capacities, extended memory, perhaps some innate predispositions for word learning, demonstrated in language-like communication systems termed as protolanguage. And given that these are properties demonstrated by trained apes after extensive learning and emerge spontaneously and with minimal experience very early in child language development and/ or as part of normal biological development, involvement of Baldwinian processes seems like a plausible explanation. It is much less likely the UG to have been targets of Baldwinian adaptation, given that these are part of linguistic behaviour of the literate adult, suggesting causal association of grammatical complexities with writing. In this sense one must remember that ever since its invention literacy and writing skills have been the privilege of a highly restricted small number of individuals until the invention of the printing press in the 1400s and the resulting mass proliferation of literacy and education. And it is not to forget that even now millions of people are still illiterate.

Alternatively, it is plausible to suggest that young brains, which come at birth with general, functionally unspecified cognitive properties, become configured for language use, following a general tendency in cognitive development where domain specific mechanisms arise during ontogeny under the influence of experience, forming domain-specific configurations of neuronal connectivity(Sherwood et all. 2008).

This suggests that the evolution of complex grammatical structure and especially multiple sentential embedding as a hallmark of language, still maintained by some linguistic paradigms, is a demonstration not of an innate UG, but a property of language, coincidentally and, likely causally related and prompted by cultural evolution and literacy and effectively applied in shaping the literate mind by encouraging self-reflection and self-improvement.

5.2. Co-evolution of the mind, culture and metacognition

This is not to say that metacognition is impossible in the absence of complex grammar, as various languageless species have demonstrated lower degrees of self-evaluation, as, as noted above, primates have demonstrated some form of metacognition by observing behaviour and estimating and anticipating actions of others. And illiterate humans are definitely not deprived of metacognitive functions.

From a different angle, one would expect that for humans living in pre-civilization environments survival in the wild does not leave time for self-reflection and analysis of mental life. Free time and attention to mental processes is made by civilization where division of labour allows for intellectual activities, some of which, do not necessarily demand, although are greatly facilitated by complex language, e.g. mental states can be discussed in dialogues with limited use of sentential recursion. That said, the exploration of mental life as a focus of attention is greatly facilitated by writing, which spurs the use of abstract vocabulary and structural sophistication. So, complex language is a good tool to have but not absolute necessity for verbalization of metacognition.

The evolution from orality to writing is a culturally driven process illustrated by the evolution of language, which shapes minds, their relation to reality, e.g. their perceptions and interactions with objects, fellow humans and the self. The mind is configured by taking advantage of the brain's biological inheritance as some innate cognitive predispositions, e.g. theory of mind, a primitive form of metacognition, episodic memory, as a phenotypic inheritance from ancestral species, and further shaped by cultural values and attitudes, reflected in language systems. By learning a language one absorbs cultural traditions and values, which molds the ways the mind processes reality, internal and external.

Attention to one's mode of thinking is instrumental to one's decision-making and learning, which is why in modern literate societies advanced metacognitive abilities are intentionally inculcated in youngsters by explicit verbal instruction. Teachers are verbal coaches helping students to notice their strengths and weaknesses in how they process information and regulate and improve their learning strategies. In this sense one could say that the Sapir-Worf hypothesis (which argues that language shapes thought), although usually interpreted as a statement that speakers of different languages think differently, is applicable here in some limited sense , i.e. using specific linguistic expressions helps one improve thinking. Moreover, specific linguistic expressions are used as a teaching tool for shaping students' thought processes. That said, some remnants of the traditional mindset, shaped by oral traditions is still present in the literate mind and demonstrated in face-to-face dialogues.

Importantly, writing and other semasiographic systems evolve as part of culture through multiple stages from cruder to ever more refined signs, e.g. beginning with a limited number of meanings, materialized by iconic signs with strong reliance on context for interpretation, gradually replacing iconic signs with symbols and reducing to minimum the reliance on context in interpretation(Croft 2017). Numerous illustrations of intermediate stages in such transformations are Sumerian script, Egyptian hieroglyphs, Chinese script, Mayan script, Aztec script. (see Ong, 2002, p.84- for details). Given that, one is prompted to speculate a similar trajectory of gradual improvement in metacognitive functions.

To sum up, the human brain is a biological inheritance. Human psychology, i.e. the human mind, its cognitive relation to outside world and to itself, including conscious manipulation of one's thoughts, i.e. metacognition, is stimulated and partially shaped by the evolution of culture and communicative technology.

5.3. Metacognition or interaction: the original function of language

There is an ongoing debate on the original function of language, where silent reflection is argued by some, especially within biolinguistic circles, attributed to the purported primordial evolution of Language of Thought (LOT) and UG predating its implementation in communication by evolution of speech. In opposition proponents of the usage-based approaches argue for communicative function as the justification of the evolution of language, given both the referential utility of the language system and its interactional use.

That said, studies reviewed above demonstrate that the communicative function of language both in ontogeny and in phylogeny predates and facilitates its metacognitive function, since actions, physiological and communicative, presuppose one's reflection on these, e.g. analyzing what one did and said, possible failure to reach a communicative goal and a plan for future improvement. Thus, communicative action precedes reflecting upon it. Moreover, the invention of wealth of linguistic forms as labels for cognitive processes of introspection and self-consciousness emerge in highly complex societies with writing traditions and literature. Thus, reflection, self-monitoring and self-evaluation is not likely to be the reason for initiating language as thought-structuring algorithm.

Summary and conclusions

Metacognition is a graded ability, available in some form to various non-human species, e.g. most notably various primate species. Given it is essential for survival, it likely has been an evolutionary target. Advanced metacognition is a feature in species of advanced intelligence. That said, in non-human species metacognition remains a matter of private thought, while humans are unique in communicating our mental life most clearly in linguistic forms.

In pre-literate cultures and literate participants of direct dialogues attention to one's mind-internal self-analysis is sidelined by predominant focus on experiences. Metacognition is stimulated by written language with the deliberate focus on one's mind and its internal deliberations, facilitated by extensive use of highly abstract linguistic forms and multilevel grammatical hierarchies of the highest order of complexity. The same pattern of interdependence of culture and language shapes the human brain and mind during development.

From evolutionary perspective it is natural to extrapolate the same pattern of interplay between language, human culture and metacognition, i.e. as language has emerged in service of direct dialogues and oral cultural traditions, the focus is on physical survival and human interaction with reality which stimulates action and leaves little opportunity for reflection and self-analysis. The emergence of complex civilizations and proliferation of writing has created the conditions for self-reflection and logical evaluation and its verbalization in language, spurring the demand for ever increasing complexity of linguistic structure. The observed overlap of metacognition and complex linguistic structure is spurred by the demands of complex civilization and is not a feature of language at its initial pre-literate stages. Thus, human culture, metacognition and language are intertwined in bio-cognitive evolution, ontogenetic development and socio-cultural transformation.

References

Ardila A. et all.(2010) Illiteracy: the neuropsychology of cognition without reading, Archives of Clinical Neuropsychology 25, 689-712, Oxford Journals, Oxford University Press, doi:10.1093/arclin/acq079
Bickerton D. **(1990**) Language and species, University of Chicago Press

Bloom P. **(2000)** How Children Learn the Meaning of Words, MIT Press
Chomsky N. (2005) Three Factors in Language Design, Linguistic Inquiry, Vol. 36, No.1, 1-22, MIT
Croft W.**(2017)** Evolutionary complexity of social cognition, semasiographic systems and language, in Mufwene, S., Coupe, C., Pellegrino, F. eds. Complexity in language, developmental and evolutionary perspectives, Cambridge University Press, chap. 5

Carruthers P., Fletcher L., Ritchie B. (2012) The evolution of self-knowledge, Philosophical topics, vol.40, no.2, p.13-37
Donald M. (1991)Origins of the modern mind, Harvard University Press
Everett D., **(2005)** Cultural Constraints on Grammar and Cognition in Piraha, Another Look at the Design Features of Human Language, Current Anthropology, vol. 46, No4, p. 621-646

Fitch T. (2010) The evolution of language, Cambridge University Press
Fugazza C., Pongracz P., Pogani A., Lenkei R., Miklosi A. (2020) Mental representation and episodic-like memory of own actions in dogs, Scientific reports,10, article number 10449
Ginsburg, J. (2001) Dynamics and semantic of the dialogue, Semantic scholar, Corpus ID 119010931

Ginsburg, J. **(2008)** Semantics for conversation, CSLI Publications

Ginsburg J. , Poesio, M., **(2016)** Grammar is a System that Characterizes Talk in Interaction , Frontiers in Psychology, vol.7, art. 1938
Givon T., **Malle B. (2002)** The evolution of language out of pre-language, John Benjamins,
Jackendoff, R.,Wittenberg, E.**(2014**) What can you say without syntax, a hierarchy of grammatical complexity, in Newmeyer, F.,Preston,L. eds. Measuring Linguistic Complexity, Oxford University Press , chap.4
Heyes C. (2012) New thinking, the evolution of human cognition, Introduction, Phyl. Trans. R. Soc. B 367, 2091-2096
Heyes C., Bang D., Shea N., Firth C., Fleming S. (2020) Knowing ourselves together: the cultural origins of metacognition, Trends in cognitive sciences, vol.4, no.5, p. 349-362, https://doi.org/10.1016/j.tics.2020.02.007
Higgins T. (2005) Humans as applied motivation scientists : self-consciousness form "shared reality and "becoming", in Terrace H.,Metcalfe J.eds. The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press, chap. 6
Jablonka ,E. Lamb M. (2005) Evolution in Four Dimensions , MIT Press,
Kinsbourne M. (2005) A continuum of self-consciousness that emerges in ontogeny and phylogeny, in Terrace H.,Metcalfe J.eds. The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press , chap.5
Kornell, N. (2009) Metacognition in humans and animals, Current directions in psychological science, vol 18, no.1, p. 11-15
Menzel C.(2005) Progress in the study of chimpanzee recall and episodic memory, in Terrace

H.,Metcalfe J.eds. The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press , chap.8
Metcalfe, J. (2008) Evolution of metacognition, in Dunlosky J., Bjork R. eds.Handbook of metamemory and memory, p. 29-46, Psychology Press
Metcalfe J. ,Schwartz B. (2016), The ghost in the machine: Self-reflective consciousness and the neuroscience of metacognition, in Dunlosky, J., Tauber S. (eds) The Oxford handbook of metamemory, p.407-424, Oxford University Press, chap.
Nelson K. (2005) Emerging levels of consciousness in early human development, in in Terrace H.,Metcalfe J.eds. The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press, chap.4
Nisbett R., Norenzayan A. (2002) Culture and cognition, in Pashler H., Medin D.L. eds. Steven'sHandbook of Experimental Psychology,vol 2: Memory and cognitive processes, Third Edition, chap. 13, John Wiley and Sons;
https://www2.psych.ubc.ca/~ara/Manuscripts/Nisbett&20%Norenzayan20%chapter.pdf
Oesch N., Dunbar R.(**2016**) The emergence of recursion in human language: mentalising predicts recursive syntax task performance , Journal of neurolinguistics xxx 1-12;
https://dx.doi.org/10.1016/j.jneuroling.2016.09.008
Ong W. (2002) Orality and literacy, Routledge
O' Madagain C., Helming KA, Schmidt M., Shupe E. Call J., Tomasello M. (2022), Great apes and human children rationally monitor their decisions. Proc.R.Soc.B 289:20222686, https://doi.org/10.1098/rspb.2021.2686
Papaleontiou-Louca E. (2008) Metacognition and Theory of Mind, Cambridge Scholars Publishing
Pinker S. **(2003**) Language evolution as adaptation to a cognitive niche, in Christiansen, Kirby, eds. Language evolution, Oxford Univ. Press, p. 16-37
Pinker S., Bloom P.(1990) Natural language and natural selection, BBS, 13 (4), p. 707-784
Qu Z., Shi L.,So BCL, Yin J.,Kwok Sc.(2023) Uncertainty monitoring and information-seeking in non-primate animals: Meta-analysis and systematic review, Front.Ethol. 2:1246370, doi:10.3389/fetho.2023.1246370
Sherwood C., Subiaul F., Zawidzki T.**(2008)** A natural history of the human mind, Tracing evolutionary changes in brain and cognition, Journal of Anatomy, Apr. 212 (4) p. 426-454, doi: 10.1111/j.1469-7580.2008.00868.x.
Slotnick S. (2017) Episodic memory in animals, Fifteen eighty four, Academic perspectives from Cambridge University Press, cambridgeblog.org
Terrace, H., Metcalfe, J.**(2005)** The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press
Terrace, H.**(2005**) Metacognition and the evolution of language, in Terrace H., Metcalfe J.eds. The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press, chap.3
Tomasello M.(2023) Social cognition and metacognition in great apes, a theory, Animal cognition, 26:25-35
Tulving E. **(2005)** Episodic memory and autonoesis, uniquely human? in Terrace H., Metcalfe J.eds. The Missing Link in Cognition, Origins of Self-Reflective Consciousness, Oxford University Press, chap.1
Zuberbuhler, K. **(2015**) Linguistic capacities of non-human animals, Wire's Cognitive Science,

vol.6.issue 3, Wiley Online, p.313-321